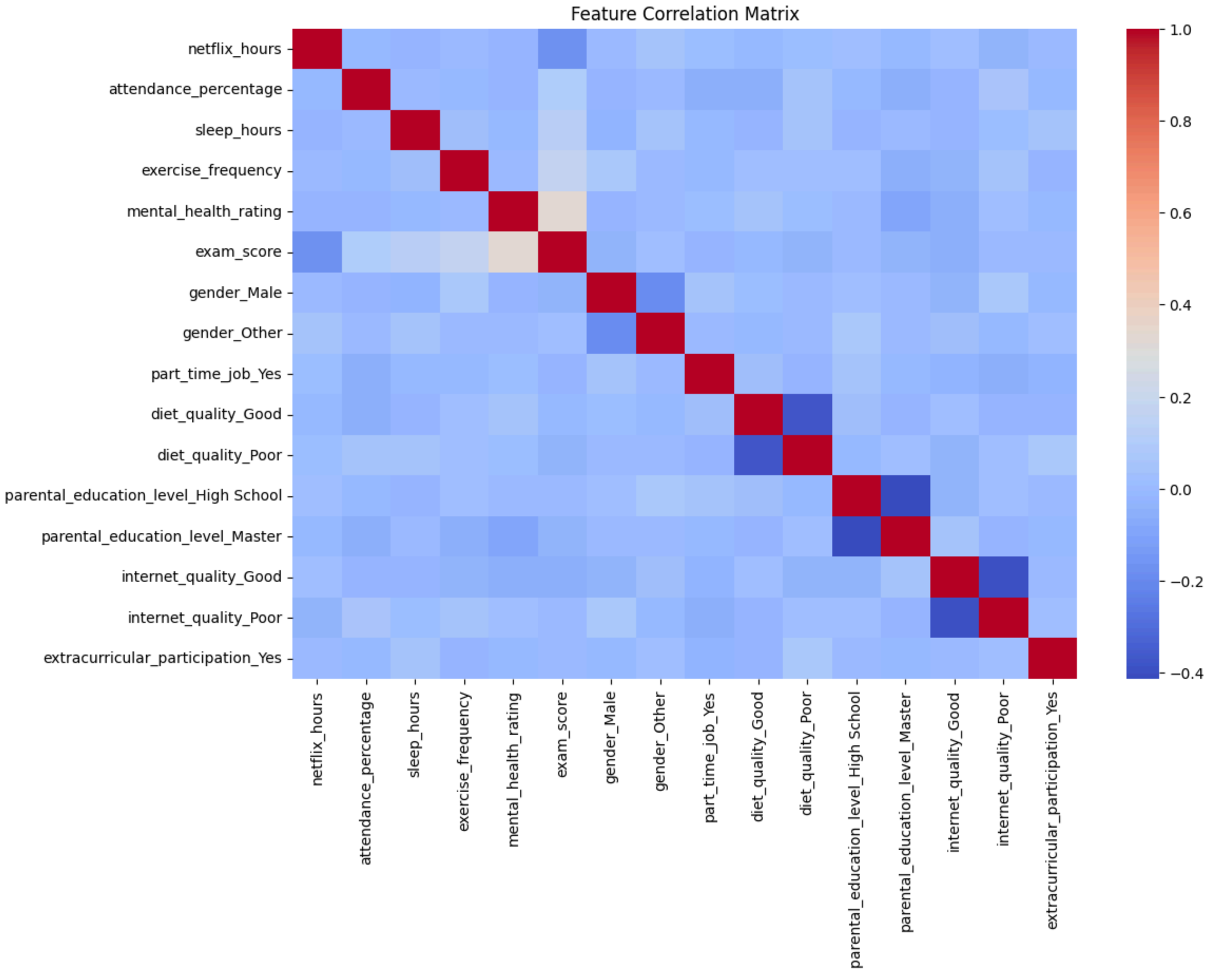
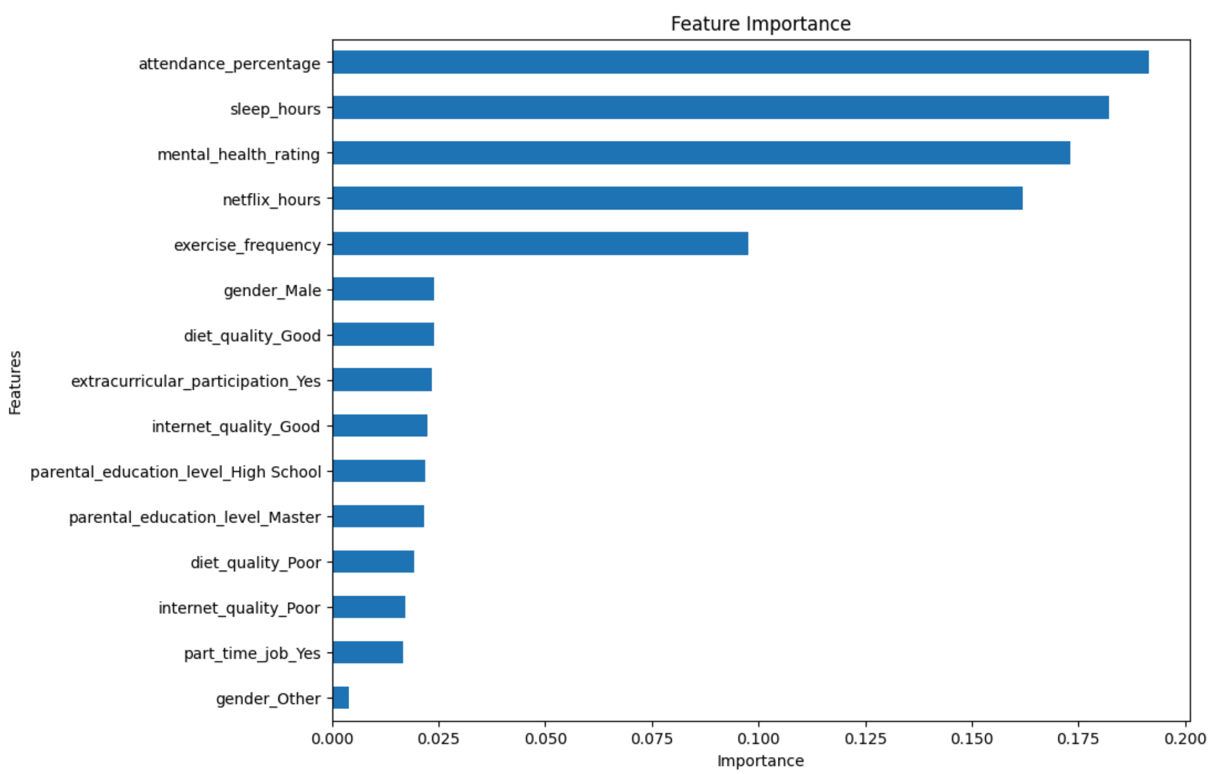
In [2]:

```
# Import libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score

# Load dataset
df = pd.read_csv("C:/Users/BALAM/Downloads/student_habits_performance.csv")

# Preview data
print(df.head())
print(df.info())

# Drop irrelevant or high-cardinality columns
df.drop(['student_id', 'age', 'study_hours_per_day', 'social_media_hours'], axis=1,

# Handle missing values
df.dropna(inplace=True)

# Encode categorical variables
df = pd.get_dummies(df, drop_first=True)

# Correlation heatmap
plt.figure(figsize=(12, 8))
sns.heatmap(df.corr(), cmap='coolwarm', annot=False)
plt.title('Feature Correlation Matrix')
plt.show()

# Define features and target
X = df.drop(['exam_score'], axis=1)
y = df['exam_score']

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_sta

# Train model
model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

# Predict and evaluate
y_pred = model.predict(X_test)
print("R² Score:", r2_score(y_test, y_pred))
print("MSE:", mean_squared_error(y_test, y_pred))

# Feature importance
importances = pd.Series(model.feature_importances_, index=X.columns)
importances.sort_values().plot(kind='barh', figsize=(10, 8), title='Feature Importa
plt.xlabel('Importance')
plt.ylabel('Features')
plt.show()
```

```
  student_id  age  gender  study_hours_per_day  social_media_hours  \
0      S1000   23  Female                  0.0                 1.2   
1      S1001   20  Female                  6.9                 2.8   
2      S1002   21    Male                  1.4                 3.1   
3      S1003   23  Female                  1.0                 3.9   
4      S1004   19  Female                  5.0                 4.4   

   netflix_hours part_time_job  attendance_percentage  sleep_hours  \
0            1.1            No                   85.0          8.0   
1            2.3            No                   97.3          4.6   
2            1.3            No                   94.8          8.0   
3            1.0            No                   71.0          9.2   
4            0.5            No                   90.9          4.9   

  diet_quality  exercise_frequency parental_education_level internet_quality  \
0         Fair                   6                   Master          Average   
1         Good                   6              High School          Average   
2         Poor                   1              High School             Poor   
3         Poor                   4                   Master             Good   
4         Fair                   3                   Master             Good   

   mental_health_rating extracurricular_participation  exam_score  
0                     8                           Yes        56.2  
1                     8                            No       100.0  
2                     1                            No        34.3  
3                     1                           Yes        26.8  
4                     1                            No        66.4  
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1000 entries, 0 to 999
Data columns (total 16 columns):
 #   Column                         Non-Null Count  Dtype  
---  ------                         --------------  -----  
 0   student_id                     1000 non-null   object 
 1   age                            1000 non-null   int64  
 2   gender                         1000 non-null   object 
 3   study_hours_per_day            1000 non-null   float64
 4   social_media_hours             1000 non-null   float64
 5   netflix_hours                  1000 non-null   float64
 6   part_time_job                  1000 non-null   object 
 7   attendance_percentage          1000 non-null   float64
 8   sleep_hours                    1000 non-null   float64
 9   diet_quality                   1000 non-null   object 
 10  exercise_frequency             1000 non-null   int64  
 11  parental_education_level       909 non-null    object 
 12  internet_quality               1000 non-null   object 
 13  mental_health_rating           1000 non-null   int64  
 14  extracurricular_participation  1000 non-null   object 
 15  exam_score                     1000 non-null   float64
dtypes: float64(6), int64(3), object(7)
memory usage: 125.1+ KB
None
```

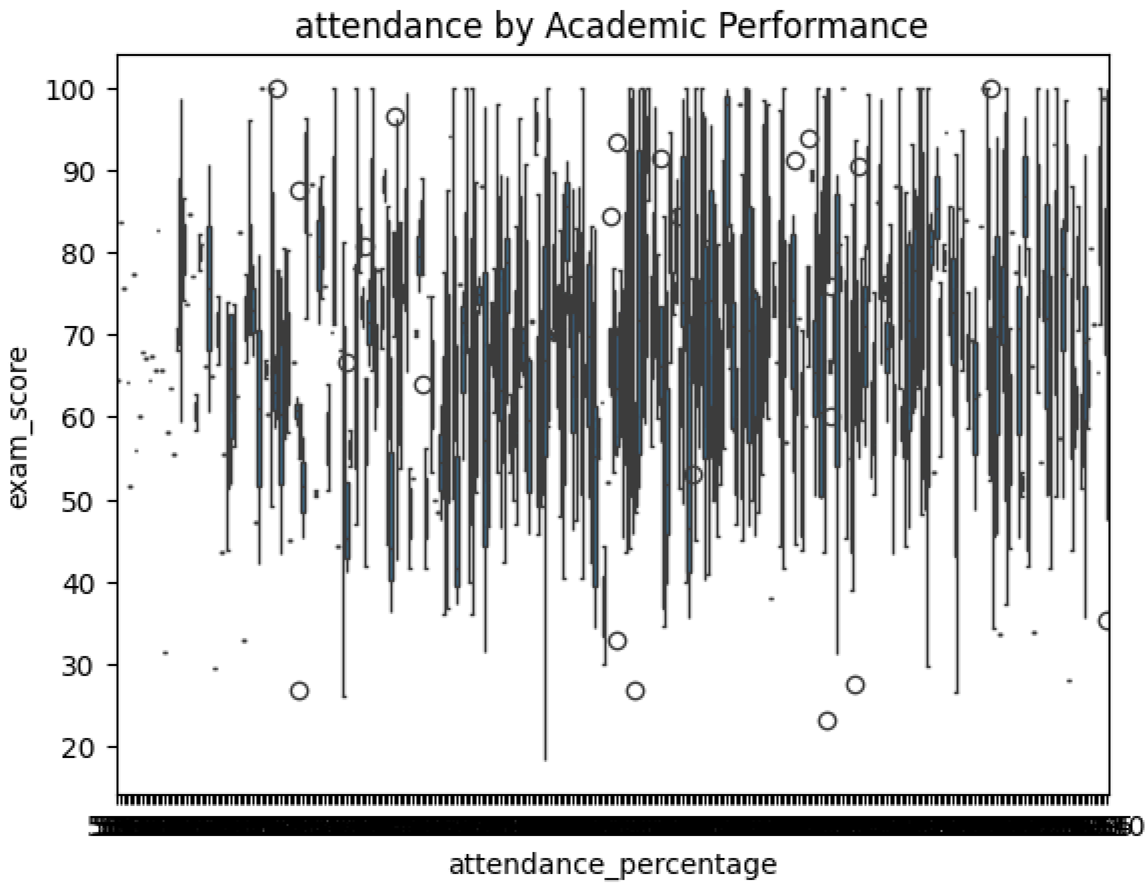



```
R² Score: 0.09201220558332879
MSE: 246.4578890494507
```



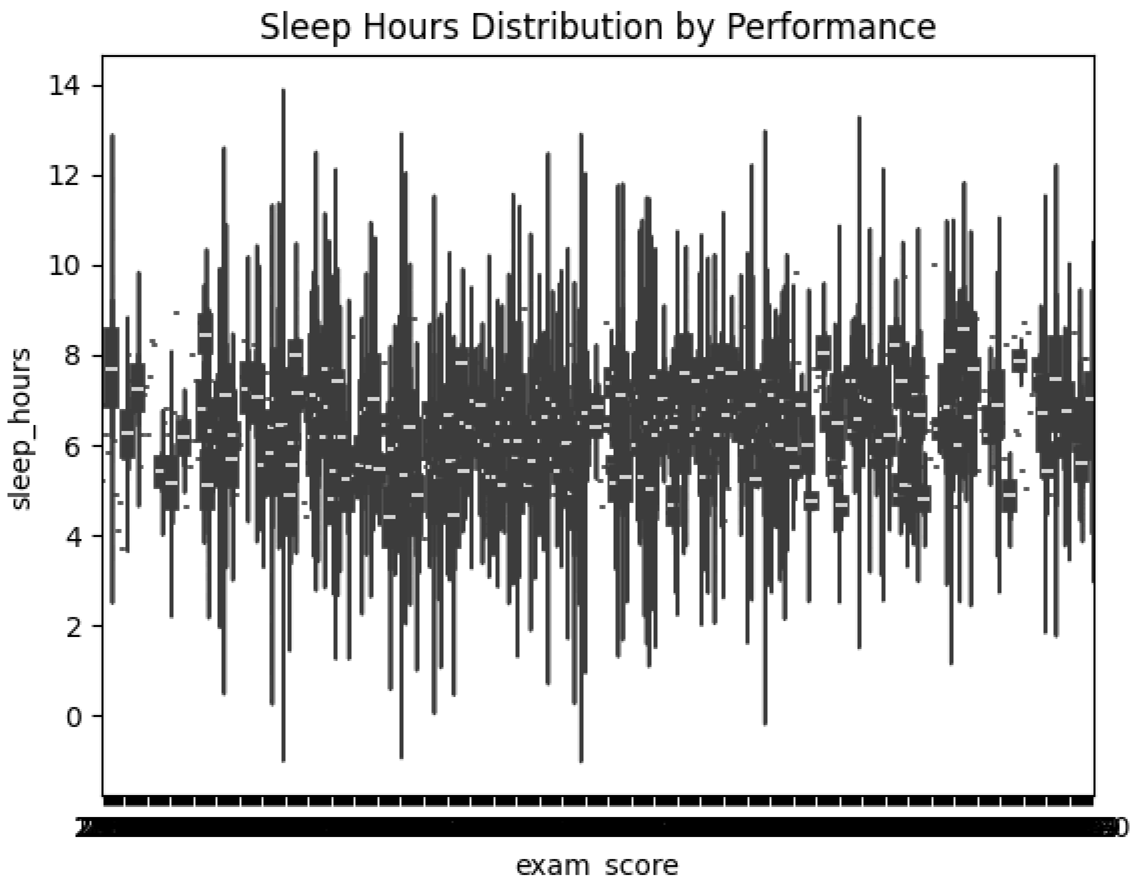
In [7]:
```
sns.boxplot(x='attendance_percentage', y='exam_score', data=df)
plt.title('attendance by Academic Performance')
```

Out[7]: Text(0.5, 1.0, 'attendance by Academic Performance')



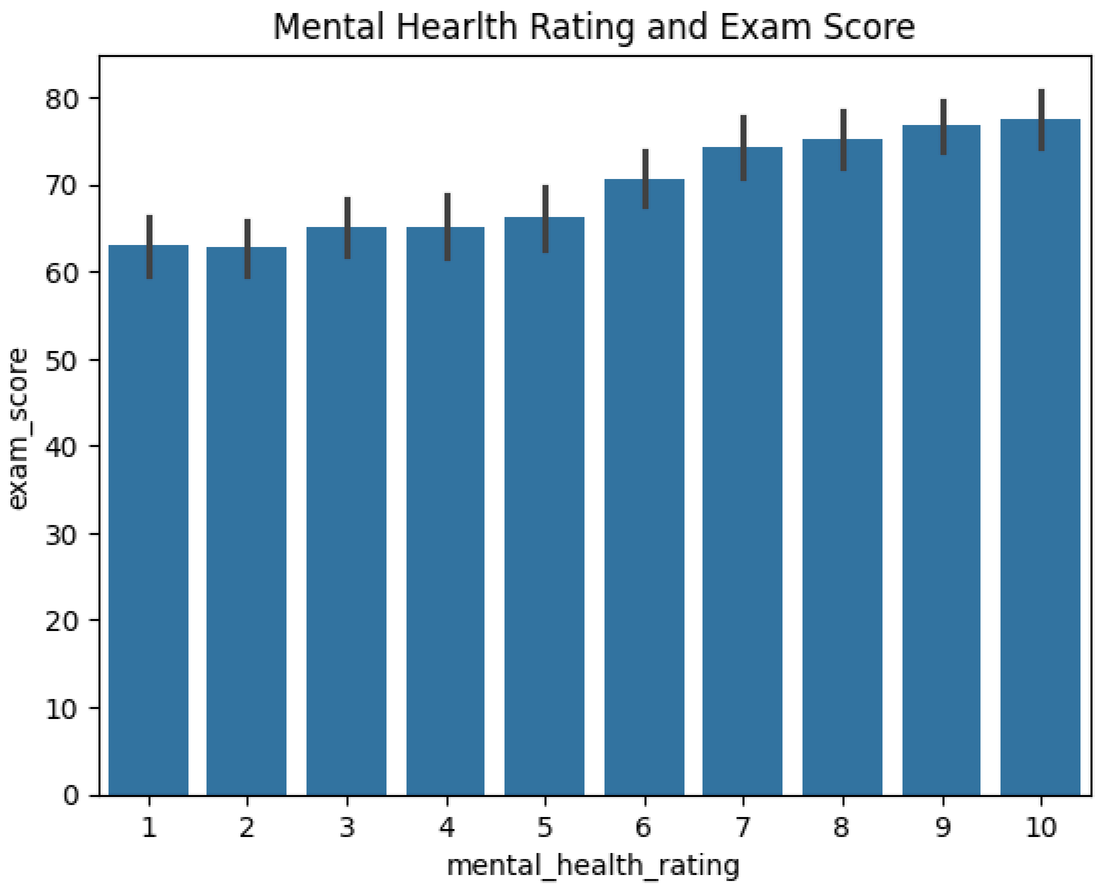
In [8]:
```
sns.violinplot(x='exam_score', y='sleep_hours', data=df)
plt.title('Sleep Hours Distribution by Performance')
```

Out[8]: Text(0.5, 1.0, 'Sleep Hours Distribution by Performance')



In [11]:
```
sns.barplot(x='mental_health_rating', y='exam_score', data=df)
plt.title('Mental Hearlth Rating and Exam Score')
```

Out[11]: `Text(0.5, 1.0, 'Mental Hearlth Rating and Exam Score')`



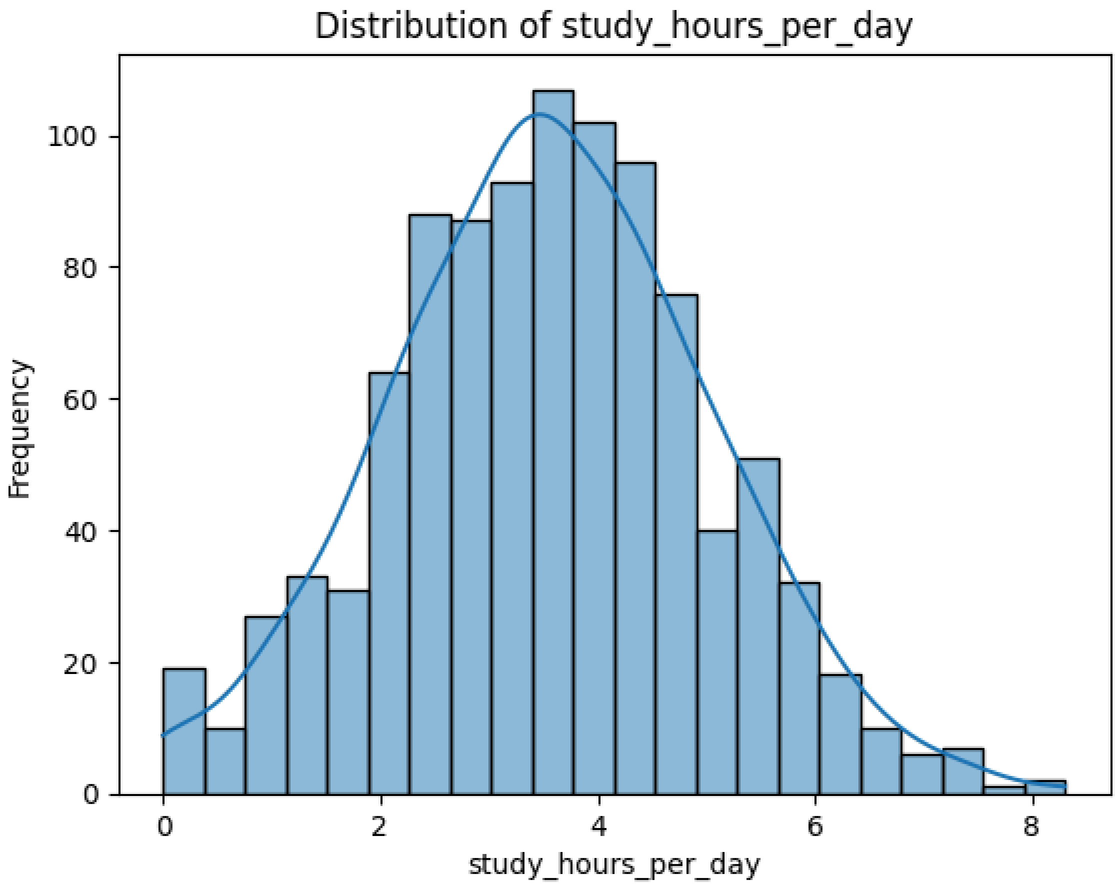
In [19]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# 2. Load dataset

# Load dataset
df = pd.read_csv("C:/Users/BALAM/Downloads/student_habits_performance.csv")
print(df.shape)
print(df.columns)
print(df.head())

# 3. Summary statistics
print(df.describe(include='all'))

# 4. Missing values
missing = df.isnull().sum()
print("Missing values:\n", missing[missing > 0])

# 5. Data types and unique values
print(df.dtypes)
for col in df.select_dtypes(include='object').columns:
    print(f"{col}: {df[col].nunique()} unique values")

# 6. Distribution plots
numeric_cols = ['study_hours_per_day', 'sleep_hours', 'social_media_hours', 'exam_s
for col in numeric_cols:
    sns.histplot(df[col], kde=True)
```

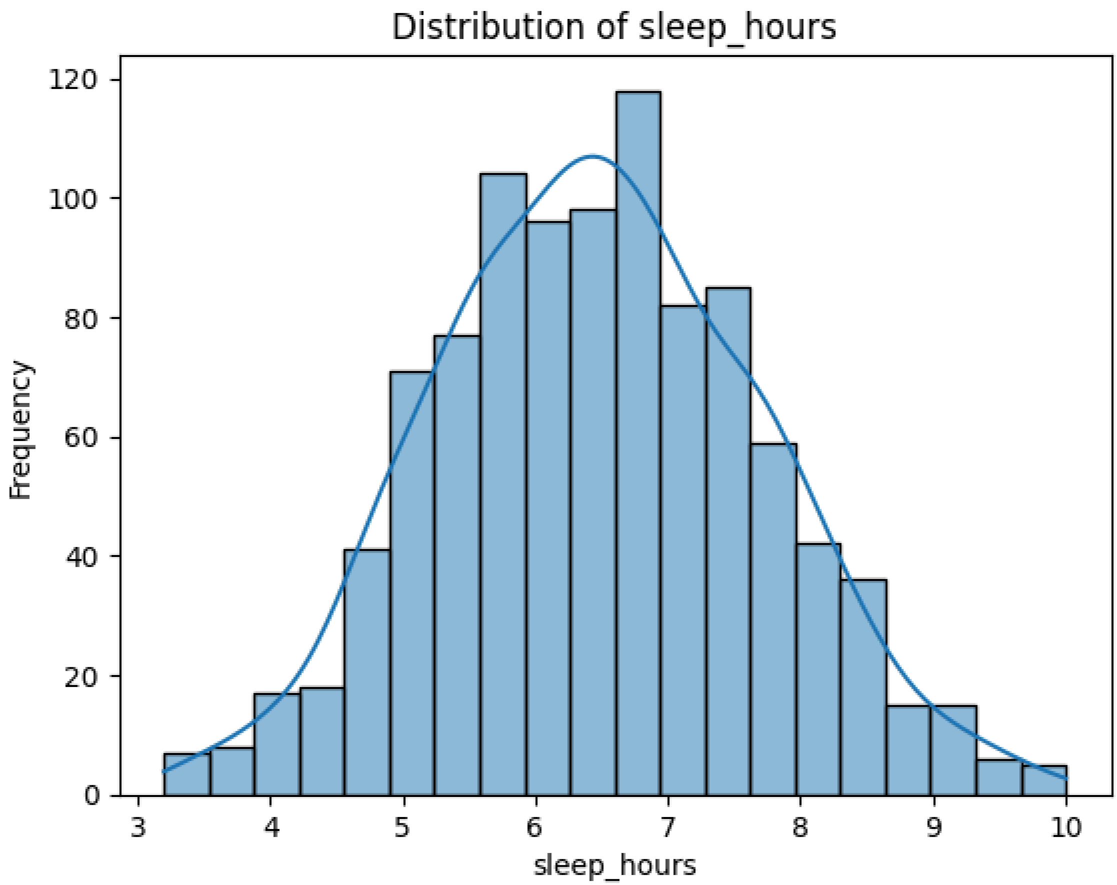
```
    plt.title(f'Distribution of {col}')
    plt.xlabel(col)
    plt.ylabel('Frequency')
    plt.show()

# 7. Boxplots for habits vs performance
sns.boxplot(x='extracurricular_participation', y='exam_score', data=df)
plt.title('extracurricular_participation vs Exam Score')
plt.show()

sns.boxplot(x='exercise_frequency', y='exam_score', data=df)
plt.title('Exercise Frequency vs Exam Score')
plt.show()

# 8. Pairplot for selected features
selected = ['study_hours_per_day', 'sleep_hours', 'social_media_hours', 'exam_score
sns.pairplot(df[selected])
plt.suptitle('Pairwise Relationships', y=1.02)
plt.show()

# 9. Grouped means
grouped = df.groupby('exercise_frequency')['exam_score'].mean()
print("Average Exam Score by exercise_frequency:\n", grouped)
```

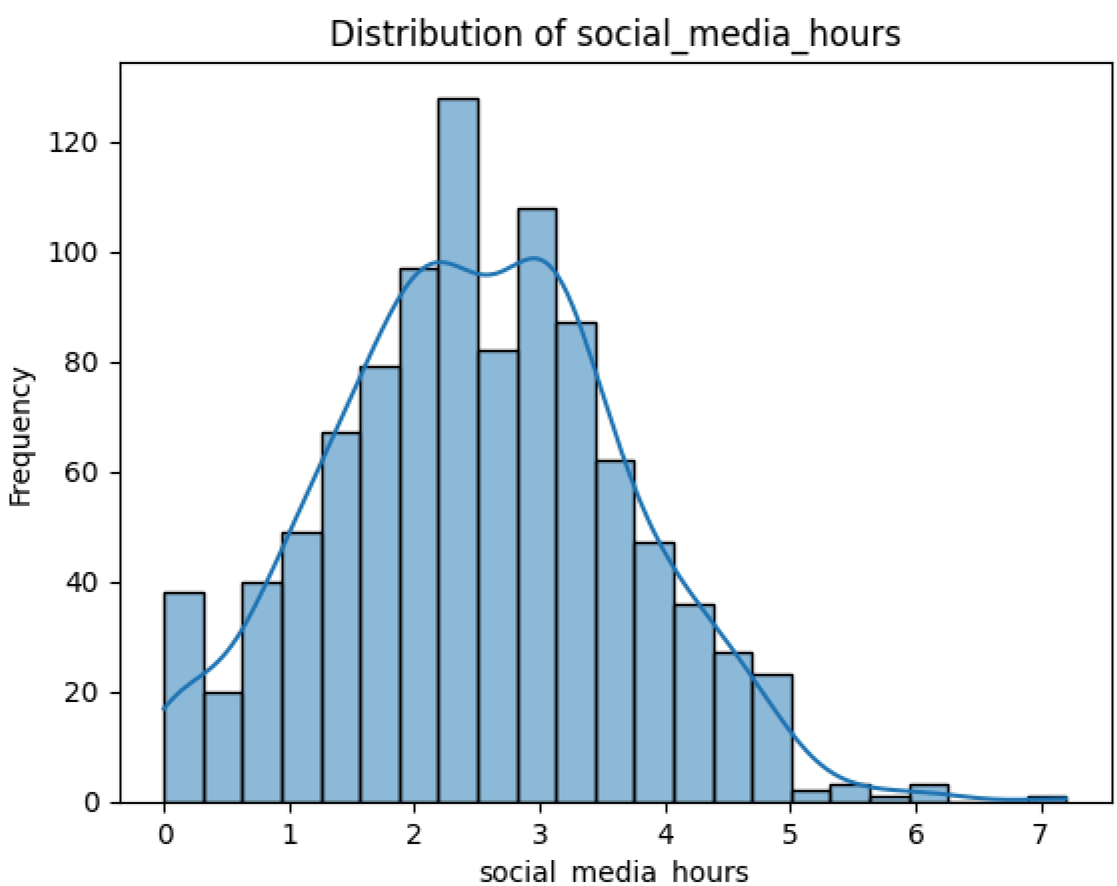
```
(1000, 16)
Index(['student_id', 'age', 'gender', 'study_hours_per_day',
       'social_media_hours', 'netflix_hours', 'part_time_job',
       'attendance_percentage', 'sleep_hours', 'diet_quality',
       'exercise_frequency', 'parental_education_level', 'internet_quality',
       'mental_health_rating', 'extracurricular_participation', 'exam_score'],
      dtype='object')
  student_id  age  gender  study_hours_per_day  social_media_hours  \
0      S1000   23  Female                  0.0                 1.2   
1      S1001   20  Female                  6.9                 2.8   
2      S1002   21    Male                  1.4                 3.1   
3      S1003   23  Female                  1.0                 3.9   
4      S1004   19  Female                  5.0                 4.4   

   netflix_hours part_time_job  attendance_percentage  sleep_hours  \
0            1.1            No                   85.0          8.0   
1            2.3            No                   97.3          4.6   
2            1.3            No                   94.8          8.0   
3            1.0            No                   71.0          9.2   
4            0.5            No                   90.9          4.9   

  diet_quality  exercise_frequency parental_education_level internet_quality  \
0         Fair                   6                   Master          Average   
1         Good                   6              High School          Average   
2         Poor                   1              High School             Poor   
3         Poor                   4                   Master             Good   
4         Fair                   3                   Master             Good   

   mental_health_rating extracurricular_participation  exam_score  
0                     8                           Yes        56.2  
1                     8                            No       100.0  
2                     1                            No        34.3  
3                     1                           Yes        26.8  
4                     1                            No        66.4  
       student_id        age  gender  study_hours_per_day  social_media_hours  \
count        1000  1000.0000    1000           1000.00000         1000.000000   
unique       1000        NaN       3                  NaN                 NaN   
top         S1999        NaN  Female                  NaN                 NaN   
freq            1        NaN     481                  NaN                 NaN   
mean          NaN    20.4980     NaN              3.55010            2.505500   
std           NaN     2.3081     NaN              1.46889            1.172422   
min           NaN    17.0000     NaN              0.00000            0.000000   
25%           NaN    18.7500     NaN              2.60000            1.700000   
50%           NaN    20.0000     NaN              3.50000            2.500000   
75%           NaN    23.0000     NaN              4.50000            3.300000   
max           NaN    24.0000     NaN              8.30000            7.200000   

        netflix_hours part_time_job  attendance_percentage  sleep_hours  \
count     1000.000000          1000            1000.000000  1000.000000   
unique            NaN             2                    NaN          NaN   
top               NaN            No                    NaN          NaN   
freq              NaN           785                    NaN          NaN   
mean         1.819700           NaN              84.131700     6.470100   
std          1.075118           NaN               9.399246     1.226377   
min          0.000000           NaN              56.000000     3.200000   
25%          1.000000           NaN              78.000000     5.600000   
```

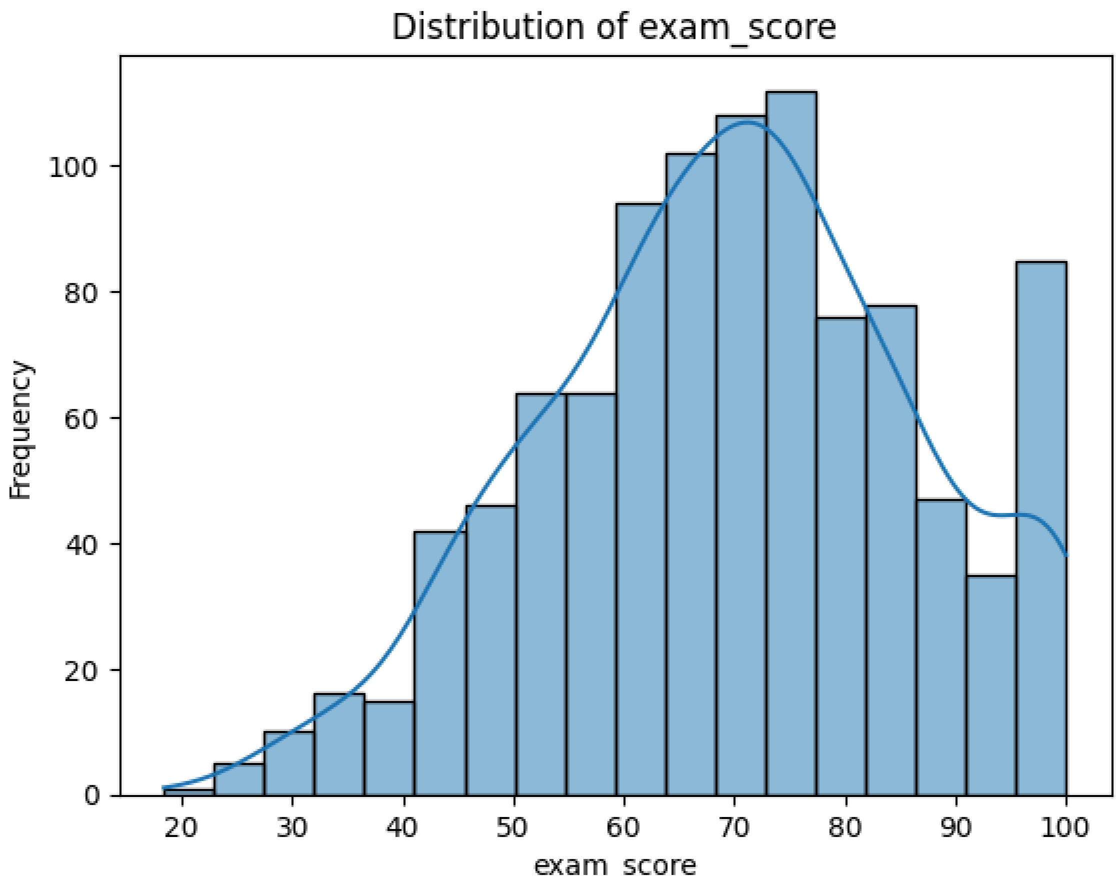
```
50%          1.800000           NaN             84.400000     6.500000
75%          2.525000           NaN             91.025000     7.300000
max          5.400000           NaN            100.000000    10.000000

       diet_quality  exercise_frequency parental_education_level  \
count          1000         1000.000000                      909
unique            3                 NaN                        3
top            Fair                 NaN              High School
freq            437                 NaN                      392
mean            NaN            3.042000                      NaN
std             NaN            2.025423                      NaN
min             NaN            0.000000                      NaN
25%             NaN            1.000000                      NaN
50%             NaN            3.000000                      NaN
75%             NaN            5.000000                      NaN
max             NaN            6.000000                      NaN

       internet_quality  mental_health_rating extracurricular_participation  \
count              1000           1000.000000                          1000
unique                3                   NaN                             2
top                Good                   NaN                            No
freq                447                   NaN                           682
mean                NaN              5.438000                           NaN
std                 NaN              2.847501                           NaN
min                 NaN              1.000000                           NaN
25%                 NaN              3.000000                           NaN
50%                 NaN              5.000000                           NaN
75%                 NaN              8.000000                           NaN
max                 NaN             10.000000                           NaN

         exam_score
count   1000.000000
unique          NaN
top             NaN
freq            NaN
mean      69.601500
std       16.888564
min       18.400000
25%       58.475000
50%       70.500000
75%       81.325000
max      100.000000
Missing values:
 parental_education_level    91
dtype: int64
student_id                        object
age                                int64
gender                            object
study_hours_per_day              float64
social_media_hours               float64
netflix_hours                    float64
part_time_job                     object
attendance_percentage            float64
sleep_hours                      float64
diet_quality                      object
exercise_frequency                 int64
```

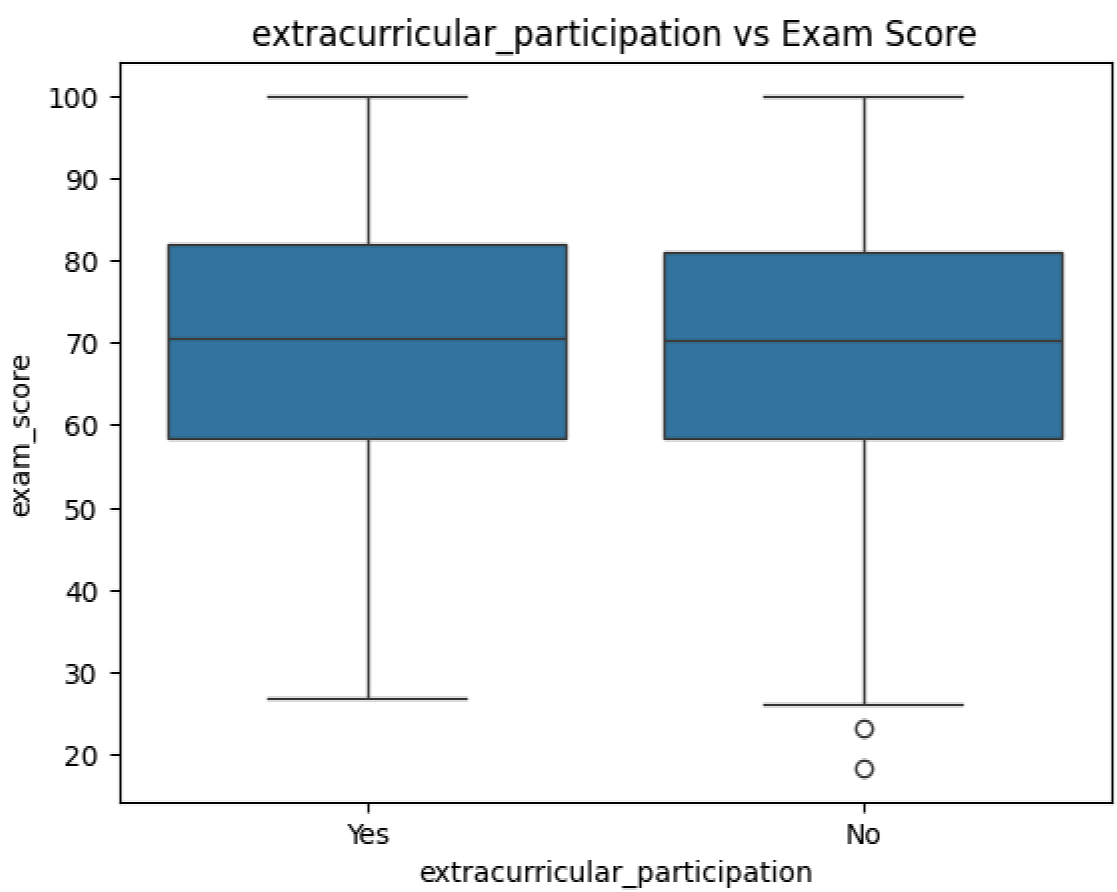
```
parental_education_level          object
internet_quality                  object
mental_health_rating               int64
extracurricular_participation     object
exam_score                       float64
dtype: object
student_id: 1000 unique values
gender: 3 unique values
part_time_job: 2 unique values
diet_quality: 3 unique values
parental_education_level: 3 unique values
internet_quality: 3 unique values
extracurricular_participation: 2 unique values
```

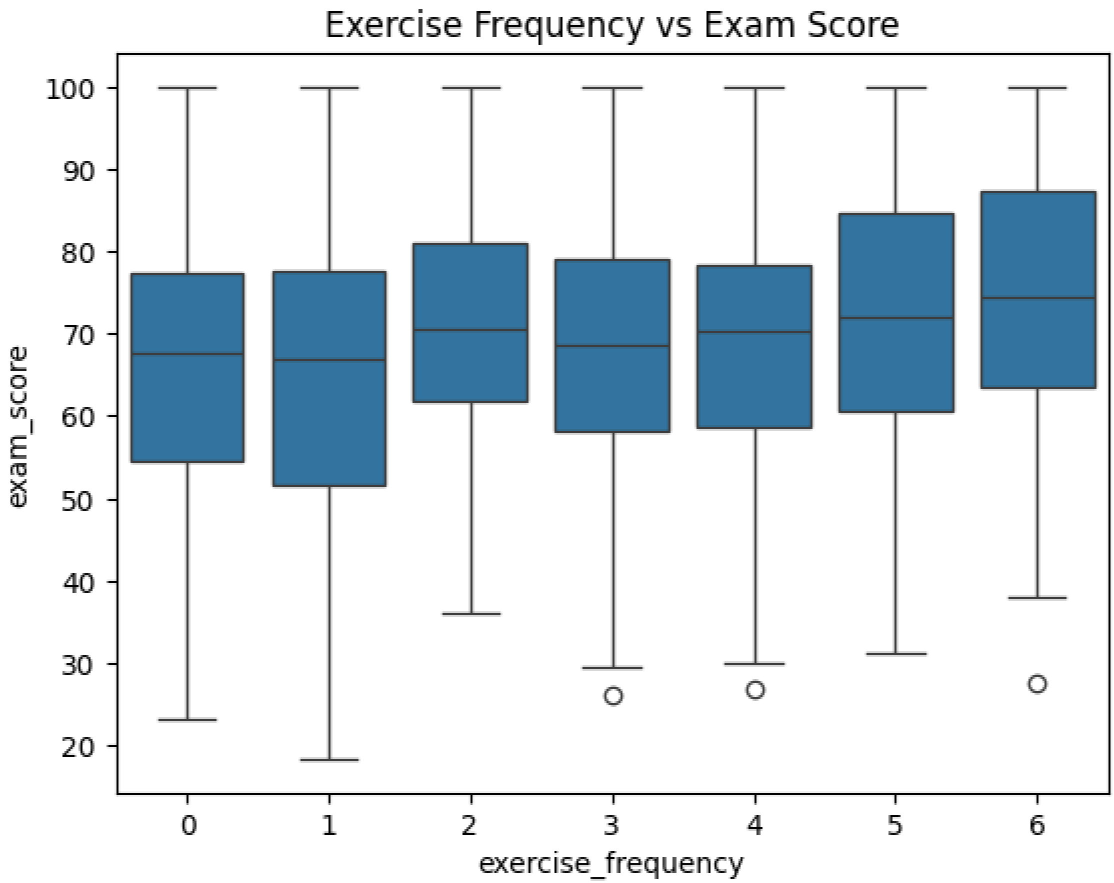


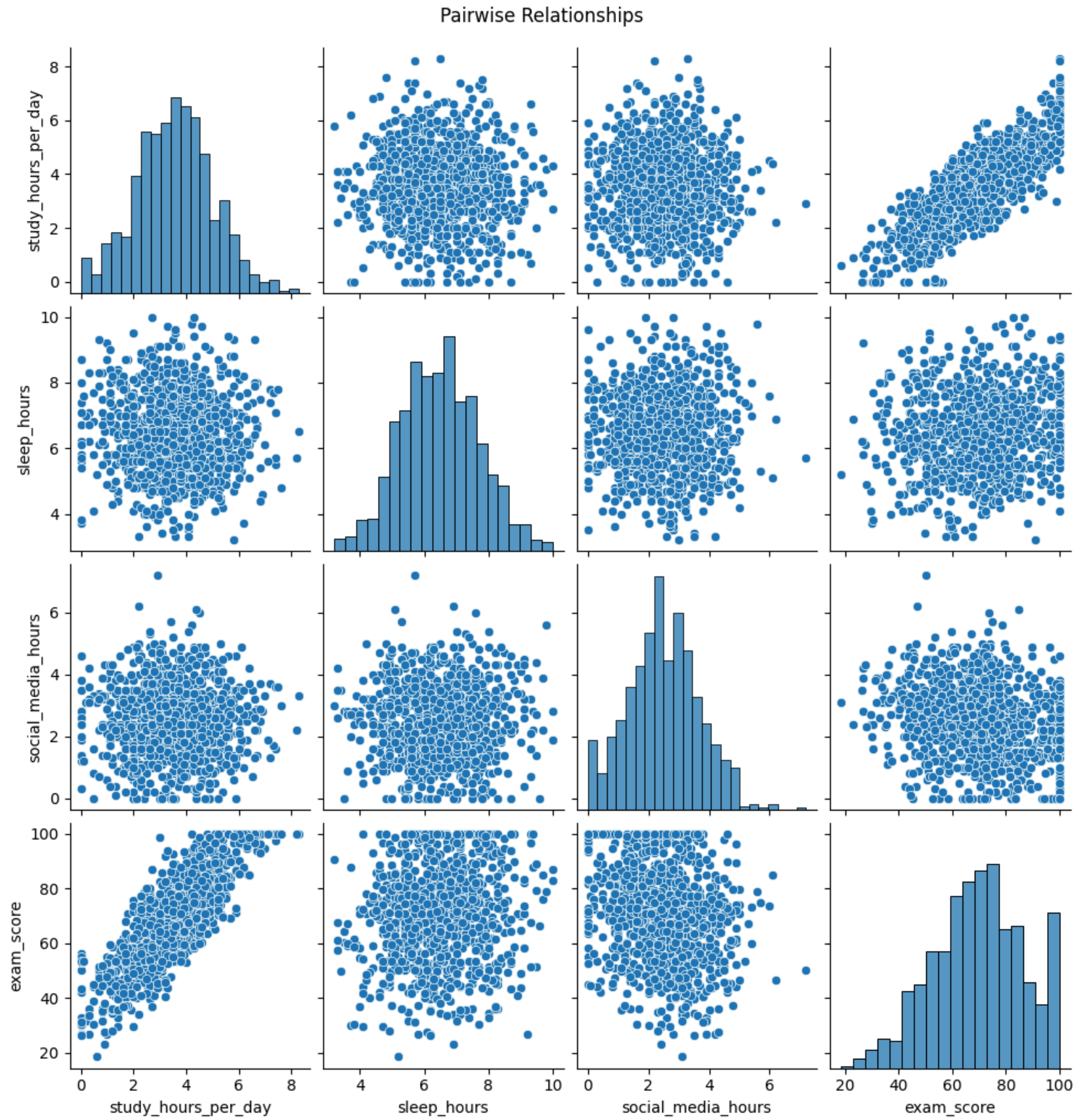

Distribution of sleep_hours
Frequency
sleep_hours
0
20
40
60
80
100
120
3
4
5
6
7
8
9
10


Distribution of social_media_hours
Frequency
social_media_hours
0
20
40
60
80
100
120
0
1
2
3
4
5
6
7


Distribution of exam_score
Frequency
0
20
40
60
80
100
20
30
40
50
60
70
80
90
100
exam_score


extracurricular_participation vs Exam Score
exam_score
20
30
40
50
60
70
80
90
100
Yes
No
extracurricular_participation


Exercise Frequency vs Exam Score
100
90
80
70
60
50
40
30
20
exam_score
0
1
2
3
4
5
6
exercise_frequency




```
Average Exam Score by exercise_frequency:
 exercise_frequency
0    66.375000
1    65.805479
2    70.459016
3    68.379085
4    68.734328
5    72.706040
6    74.567763
Name: exam_score, dtype: float64
```